

TUCK CONSULTING CLUB CASE BOOK

2006-2007 Edition



Disclaimer: This case book is a collection of contributions made by student members and friends of the Tuck Consulting Club. It does not represent the views or opinions of the Tuck Consulting Club or the Tuck School of Business at Dartmouth.
© Tuck Consulting Club, 2006

– NOT FOR SALE – NOT FOR DISTRIBUTION –

TUCK CONSULTING CLUB CASE BOOK

2006-2007 Edition

1 9 0 0

Disclaimer: This case book is a collection of contributions made by student members and friends of the Tuck Consulting Club. It does not represent the views or opinions of the Tuck Consulting Club or the Tuck School of Business at Dartmouth. © Tuck Consulting Club, 2006

– NOT FOR SALE – NOT FOR DISTRIBUTION –

TUCK CONSULTING CLUB CASEBOOK

TABLE OF CONTENTS

I. INTRODUCTION	3
II. RECRUITING INTELLIGENCE	4
Interview process timeline	4
What are the firms looking for?	5
Structure of a case interview (example).....	7
How to give a case interview	7
III. STRATEGIC TOOLS AND FRAMEWORKS.....	9
General Tools for Case Interviews	9
General Strategy.....	10
Star Model.....	10
SWOT Analysis	10
BCG Matrix	11
Porter's Five Forces	12
Core Competency Analysis.....	12
GE / McKinsey Matrix.....	13
Decision Trees	13
Marketing / Positioning.....	15
The Three Cs (Or Is It 7?).....	15
The 4 Ps.....	15
Segmentation.....	16
Operations / Productivity	17
Framework for Operations Strategy.....	17
Value Chain Analysis	17
Vertical Integration	18
Synergies.....	18
Product/Technology Life Cycle	18
Just-in-Time Production.....	19
Total Quality Management (TQM).....	19
Financial Analysis / Profitability	20
Profit Tree	20
Cost Driver Analysis.....	20
Market Sizing.....	21
Breakeven Point.....	21
Additional Resources	22
IV. PRACTICE CASES.....	23
Chateau	23
Bun on the Run	28
Switching Costs	31
Chicken Pox	34
Credit Card Collections.....	40
The End of the Road	43
Everlasting Light Bulb	46

TABLE OF CONTENTS

I. INTRODUCTION	3
II. RECRUITING INTELLIGENCE	4
Interview process timeline	4
What are the firms looking for?.....	5
Structure of a case interview (example).....	7
How to give a case interview.....	7
III. STRATEGIC TOOLS AND FRAMEWORKS.....	9
General Tools for Case Interviews	9
General Strategy.....	10
Star Model.....	10
SWOT Analysis	10
BCG Matrix	10
Porter's Five Forces	11
Core Competency Analysis.....	12
GE / McKinsey Matrix.....	12
Decision Trees	13
Marketing / Positioning.....	13
The Three Cs (Or Is It 7?).....	15
The 4 Ps.....	15
Segmentation.....	16
Operations / Productivity	17
Framework for Operations Strategy.....	17
Value Chain Analysis	17
Vertical Integration	17
Synergies.....	18
Product/Technology Life Cycle.....	18
Just-in-Time Production.....	19
Total Quality Management (TQM).....	19
Financial Analysis / Profitability	20
Profit Tree	20
Cost Driver Analysis.....	20
Market Sizing.....	21
Breakeven Point	21
Additional Resources	22
IV. PRACTICE CASES.....	23
Chateau	23
Bun on the Run	23

.....	28 Switching Costs
.....	31 Chicken Pox
.....	34 Credit Card
Collections.....	40 The End of the Road
.....	43 Everlasting Light
Bulb.....	46

Fine China	50
Got Gas?.....	55
What's a bar worth?	58
Only you can prevent accidents!.....	61
Kicks	66
Rafting.....	75
Snap, Crackle, Pop.....	79
Splat!	83
Go to China	88
Home Security Systems – The Right Move?	92
Magazine Market	96
New York Taxi Driver	99

Gas?.....	55	What's a bar
worth?	58	Only you can prevent
accidents!.....	61	Kicks
.....		66
Rafting.....	75	Snap,
Crackle, Pop.....	79	Splat!
.....	83	Go to China
.....	88	Home Security
Systems – The Right Move?.....	92	Magazine Market
.....	96	New York Taxi Driver
.....	99	

I. INTRODUCTION

This book provides a brief overview of the case interview process, a few tools to keep in mind when preparing for interviews and a collection of practice cases. We hope it is helpful in your preparations. Thank you to all our contributors:

Ted Callahan T'07
Pete Czerepak T'04
Mark Daoust T'07
Bill Dering T'05
Benjamin Farmer T'07
Elizabeth Gaydos T'07
Michael Guido T'07
Mark Hanrahan T'07
Shawn Kelly T'07
Ben King T'07
Matt Kummell T'06
Rob Langrick T'06
Mike Lawlor T'06
Michael Lewis T'06
Alejandro Lizcano T'07
David Malek T'07
Shikun Pan T'07
Tony Portera T'07
Ravi Raj I. Prabakaran T'07
Becky Rice
Daniel Rinehart T'06
Michael Sinclair T'07
Mike Song T'06
Whitney Stull T'07
Greg Wilson T'06
Karl Zimmermann T'07

GOOD LUCK

This book provides a brief overview of the case interview process, a few tools to keep in mind when preparing for interviews and a collection of practice cases. We hope it is helpful in your preparations. Thank you to all our contributors:

Ted Callahan T'07 Pete Czerepak T'04 Mark Daoust T'07 Bill Dering T'05 Benjamin Farmer T'07 Elizabeth Gaydos T'07 Michael Guido T'07 Mark Hanrahan T'07 Shawn Kelly T'07 Ben King T'07 Matt Kummell T'06 Rob Langrick T'06 Mike Lawlor T'06 Michael Lewis T'06 Alejandro Lizcano T'07 David Malek T'07 Shikun Pan T'07 Tony Portera T'07 Ravi Raj I. Prabakaran T'07 Becky Rice Daniel Rinehart T'06 Michael Sinclair T'07 Mike Song T'06 Whitney Stull T'07 Greg Wilson T'06 Karl Zimmermann T'07

GOOD LUCK

© Tuck Consulting Club, 2006 Page 3

II. RECRUITING INTELLIGENCE

Interview process timeline

- Over the summer
 - **Prepare your resume:** Think of succinct ways to describe your main achievements. Try to show the skills consultants are looking for (leadership, teamwork, academic excellence, analytical skills, problem solving, personal achievement and personal impact).
 - **Read the newspaper:** Use current events to practice breaking down business problems and thinking about how to structure a solution.
- First few weeks at school
 - **Participate in resume review:** Leverage CDO student fellows, second years and study group members to refine and revise your existing resume.
 - **Attend Sector Smarts:** Attend panels of recent Tuck alumni who share their specific industry experiences (panels range from consulting to banking, to non-profit to general management). Ask questions and decide what is important to you in your choice of career. Decide whether consulting meets your objectives.
- September
 - **Attend company briefings:** Attend as many briefings as you can in your area of interest. Many of the firms that recruit at Tuck will come to Hanover to make formal presentations. These presentations are often followed by an informal Q&A period which gives you the chance to meet some of the firm's employees. (Note - many of those presenting will be Tuck alumni and may well be involved with interviewing later in the process – first impressions matter). Try to learn about the culture of each firm and the values it holds.. Do your best to understand how the firms differ.
 - **Talk to second years:** The recruiting period can be bewildering, so talk to those that have been through it already. This is the best place to ask questions that you might be embarrassed to ask elsewhere. Remember that most second years will not be around after Thanksgiving.
- October
 - **Attend Crack the Case workshops:** These workshops are hosted by second years with prior or internship experience with management consulting firms. These four in depth workshops, each building on the previous, are designed to help you perform your best in the case interviews used by many firms.
 - **Partake in Boston/New York visit:** This three-day trip over the October break provides a comparative look at all the major firms. We visit each

Interview process timeline

- Over the summer

- o Prepare your resume: Think of succinct ways to describe your main achievements. Try to show the skills consultants are looking for (leadership, teamwork, academic excellence, analytical skills, problem solving, personal achievement and personal impact).
- o Read the newspaper: Use current events to practice breaking down business problems and thinking about how to structure a solution.

- First few weeks at school

- o Participate in resume review: Leverage CDO student fellows, second

- years and study group members to refine and revise your existing resume.
- o Attend Sector Smarts: Attend panels of recent Tuck alumni who share their specific industry experiences (panels range from consulting to banking, to non-profit to general management). Ask questions and decide what is important to you in your choice of career. Decide whether consulting meets your objectives.

- September

- o Attend company briefings: Attend as many briefings as you can in your

- area of interest. Many of the firms that recruit at Tuck will come to Hanover to make formal presentations. These presentations are often followed by an informal Q&A period which gives you the chance to meet some of the firm's employees. (Note - many of those presenting will be Tuck alumni and may well be involved with interviewing later in the process – first impressions matter). Try to learn about the culture of each firm and the values it holds.. Do your best to understand how the firms differ.
- o Talk to second years: The recruiting period can be bewildering, so talk to

- those that have been through it already. This is the best place to ask questions that you might be embarrassed to ask elsewhere. Remember that most second years will not be around after Thanksgiving.

- October

- o Attend Crack the Case workshops: These workshops are hosted by second years with prior or internship experience with management consulting firms. These four in depth workshops, each building on the previous, are designed to help you perform your best in the case interviews used by many firms.

- o Partake in Boston/New York visit: This three-day trip over the October

- break provides a comparative look at all the major firms. We visit each

firm at their home office – another opportunity to find the firms with the best fit for you.

- **Establish links with Tuck Alumni:** Building relationships at your firms of choice will express your interest and help develop your understanding of firms, particularly if you have a strong location preference
 - **Practice case interviews:** Work with your peers (both friends and other you are not close with) and work by yourself (you don't need an interviewer to practice structuring a problem or performing "public math")
- November/December
 - **Apply to firms:** Deadlines and methods vary for each firm. If in doubt, then ask. You should also have a back up plan in case you are unsuccessful with your first choice career. Tuck's policy is that 50% of interview slots are "closed list" positions. A "closed list" position means you were selected by the firm for an interview. If you are not "closed listed" you can still bid for an interview slot.
 - **Practice case interviews:** Continue to work with your peers and work by yourself. Some firms may visit campus and run sessions to help you prepare for the interviews
 - January
 - **Refine technique:** Formal second year practice coaching session will be organized with second years (and in some visiting firms) This period is very busy and involves a lot of balancing of school work and interview preparation. **Prioritize your practice.**
 - February
 - **Formal on campus interviews.** First round interviews are typically held on campus. Firm's interview practices are similar but key differences do exist. Second round interviews will usually be at the firm's office. Again, the exact process varies from firm to firm. – Speak with the CDO if you have any questions.

The entire process is well tried and tested. You will get a lot of help but it also requires a concerted effort on your part to be successful. Don't underestimate the importance of practicing cases.

What are the firms looking for?

Key success factors associated with each firm will vary, however there are a number common points to remember when compiling your resume, submitting applications, and interviewing. These key success factors are divided into two key areas: 1) case-based skills and 2) Experienced based characteristics:

with Tuck Alumni: Building relationships at your firms

of choice will express your interest and help develop your understanding of firms, particularly if you have a strong location preference

- o Practice case interviews: Work with your peers (both friends and other you are not close with) and work by yourself (you don't need an interviewer to practice structuring a problem or performing "public math")

- November/December

- o Apply to firms: Deadlines and methods vary for each firm. If in doubt, then ask. You should also have a back up plan in case you are unsuccessful with your first choice career. Tuck's policy is that 50% of interview slots are "closed list" positions. A "closed list" position means you were selected by the firm for an interview. If you are not "closed listed" you can still bid for an interview slot.
- o Practice case interviews: Continue to work with your peers and work by yourself. Some firms may visit campus and run sessions to help you prepare for the interviews

- January

- o Refine technique: Formal second year practice coaching session will be organized with second years (and in some visiting firms) This period is very busy and involves a lot of balancing of school work and interview preparation. Prioritize your practice.

- February

- o Formal on campus interviews. First round interviews are typically held on campus. Firm's interview practices are similar but key differences do exist. Second round interviews will usually be at the firm's office. Again, the exact process varies from firm to firm. – Speak with the CDO if you have any questions.

The entire process is well tried and tested. You will get a lot of help but it also requires a concerted effort on your part to be successful. Don't underestimate the importance of practicing cases.

What are the firms looking for?

Key success factors associated with each firm will vary, however there are a number common points to remember when compiling your resume, submitting applications, and interviewing. These key success factors are divided into two key areas: 1) case-based skills and 2) Experienced based characteristics:

Case Based Skills

- **Problem solving skills (dealing with ambiguity):** These themes may seem obvious when considering the case interview but are mentioned here for two reasons. First, does your resume show a history of success in this area? Secondly, don't let yourself get flustered by interviews that don't follow the typical path. Stick to a logical approach even in less structured interviews.
- **Analytical Skills:** Your analytical skills are often tested through your ability to structure a problem and its various components. This skill can be learned through practice and refined through additional practice. It is important to have structure in your problem solving as you will need to effectively communicate your plan of action to tackle the problem. A well designed structure makes this communication easy and will help you effectively solve the problem. Do not confuse structure with frameworks (the frameworks provided in this book will help you think about an appropriate structure) but they are seldom used in isolation as a structure to solve the problem.
- **Quantitative skills:** Most cases will involve a quantitative component. These math problems typically involve high school level math. To complicate the experience, you are often not given all required information required to solve the problem. Your setup is equally or more important to your ability to use solve basic math. Also remember to walk your interviewer through your calculations and approach as you complete the problem vs. being a silent calculator.
- **Syntheses skills:** Your ability to weave insights together throughout the case and to develop a coherent synthesis (not a summary or travel log) is important in any case interview.

Experience based characteristics

- **Personal Achievement:** You will be up against a competitive group of applicants that all have similarly impressive lists of academic and professional achievements. Distinguish yourself by showing that you are equally successful in your personal life by using examples from outside work/school whenever possible. This will add a breadth and interest to your candidacy.
- **Leadership:** Discussion of leadership is a common activity at Tuck but it can be hard to quantify. Some people will have obvious leadership experience but for others, you will need to make sure that leadership elements in past experiences stand out. Practice describing these events in a succinct and understandable way.
- **Teamwork:** Some would call this the "airport test." Are you someone that the interviewer wants to work with on a difficult and time consuming case or be stuck in an airport with? The interview is your chance to show that you fit with the culture of your chosen firm/s. This is a two way process: do they like you and do you like them? The best advice here is to try to relax and be yourself.

- Problem solving skills (dealing with ambiguity): These themes may seem obvious when considering the case interview but are mentioned here for two reasons. First, does your resume show a history of success in this area? Secondly, don't let yourself get flustered by interviews that don't follow the typical path. Stick to a logical approach even in less structured interviews.

- Analytical Skills: Your analytical skills are often tested through your ability to structure a problem and its various components. This skill can be learned through practice and refined through additional practice. It is important to have structure in your problem solving as you will need to effectively communicate your plan of action to tackle the problem. A well designed structure makes this communication easy and will help you effectively solve the problem. Do not confuse structure with frameworks (the frameworks provided in this book will help you think about an appropriate structure) but they are seldom used in isolation as a structure to solve the problem.

- Quantitative skills: Most cases will involve a quantitative component. These math problems typically involve high school level math. To complicate the experience, you are often not given all required information required to solve the problem. Your setup is equally or more important to your ability to use solve basic math. Also remember to walk your interviewer through your calculations and approach as you complete the problem vs. being a silent calculator.

- Syntheses skills: Your ability to weave insights together throughout the case and to develop a coherent synthesis (not a summary or travel log) is important in any case interview.

Experience based characteristics

- Personal Achievement: You will be up against a competitive group of applicants that all have similarly impressive lists of academic and professional achievements. Distinguish yourself by showing that you are equally successful in your personal life by using examples from outside work/school whenever possible. This will add a breadth and interest to your candidacy.

- Leadership: Discussion of leadership is a common activity at Tuck but it can be hard to quantify. Some people will have obvious leadership experience but for others, you will need to make sure that leadership elements in past experiences stand out. Practice describing these events in a succinct and understandable way.

- Teamwork: Some would call this the "airport test." Are you someone that the interviewer wants to work with on a difficult and time consuming case or be stuck in an airport with? The interview is your chance to show that you fit with the culture of your chosen firm/s. This is a two way process: do they like you and do you like them? The best advice here is to try to relax and be yourself.

There are many other areas that could be mentioned. Some firms will rate some higher than others. Try to build a picture of this as you listen to firm presentations, talk to employees and scour the respective firm websites.

Structure of a case interview *(example)*

Order	Section	Duration	Skills	Objective
1	Meet & greet	1-5 min	Inter-personal	- Utilize information from the interviewer bio to make a connection - Relate to the interviewer with similar experiences (from Tuck, if possible)
2	Receive the question	1-2 min	Listening	- Take notes - Pick up as much data as possible
3	Clarify the question	2-3 min	Finding data	- Rephrase the question - Dig a little deeper with follow-up questions with general strategy frameworks (e.g., 4 P's/3 C's)
4	Develop a framework to solve	10-15 min	Analytics, problem solving, and public math	- Frame problem - Make hypotheses - Walk the interviewer through the solution
5	State the conclusion	1-2 min	Big picture thinking; synthesis	- Synthesize findings - State larger considerations of the solution (overall impact)

How to give a case interview

Your role as a case interviewer is to present a problem and act as guide for the interviewee. The interviewee does most of the work and the best interviewer will stay mostly passive with limited directive probing throughout the process. Here are some interviewer best practices:

- **Prepare:** Make sure you have read the case before the practice session. You should understand the most important facts before you sit down with your partner.
 - All too frequently an interviewee is shortchanged by an interviewer who is not able to adequately facilitate
- **Role Play / Facilitate:** Act as a consultant for interviewee's desired firm

There are many other areas that could be mentioned. Some firms will rate some higher than others. Try

to build a picture of this as you listen to firm presentations, talk to employees and scour the respective firm websites.

Structure of a case interview (example)

Order Section Duration Skills Objective

1 Meet & greet 1-5 min Inter-personal • Utilize information from the interviewer bio to make a connection

• Relate to the interviewer with similar experiences (from Tuck, if possible) 2 Receive the question

1-2 min Listening • Take notes

• Pick up as much data as possible 3 Clarify the question

2-3 min Finding data • Rephrase the question

• Dig a little deeper with follow-up questions with general strategy frameworks (e.g., 4 P's/3 C's) 4 Develop a

framework to solve

10-15 min Analytics, problem solving, and public math

• Frame problem

• Make hypotheses

• Walk the interviewer through the solution 5 State the conclusion

1-2 min Big picture

thinking; synthesis

• Synthesize findings

• State larger considerations of the solution (overall impact)

How to give a case interview

Your role as a case interviewer is to present a problem and act as guide for the interviewee. The interviewee does most of the work and the best interviewer will stay mostly passive with limited directive probing throughout the process. Here are some interviewer best practices:

- Prepare: Make sure you have read the case before the practice session. You should understand the most important facts before you sit down with your partner.

- o All too frequently an interviewee is shortchanged by an interviewer who is not able to adequately facilitate

- Role Play / Facilitate: Act as a consultant for interviewee's desired firm

- Set the scene with small talk
- Control the time of the interview through the questions you pose, shoot for about 25 minutes in length
- Selectively challenge the interviewee's hypotheses when appropriate to probe deeper and better analysis.
- Be tough, but give subtle clues when at an impasse
- **Be complete:** Make this a real interview experience
 - Make the interviewee synthesize the key points
 - Ask resume or personal background questions
 - Allow the interviewee to practice their questions to the interviewer
- **Respond:** Take notes and give feedback at the end, so that the interviewee can learn from the process. This feedback should:
 - Clarify what went well and not so well
 - Present elements of the case that were missed
 - Direct the interviewee toward areas for improvement
 - Be comprehensive as you address the interviewee's presences and poise in addition to the technical skills they present

Note to all interviewers: Always give critical and constructive feedback. Be as honest and forthcoming as will be meaningful to your interviewee. Mutual improvement is your goal and may require uncomfortable conversations.

for

about 25 minutes in length o Selectively challenge the interviewee's hypotheses when appropriate to probe deeper and better analysis. o Be tough, but give subtle clues when at an impasse

- Be complete: Make this a real interview experience

- o Make the interviewee synthesize the key points
 - o Ask resume or personal background questions
 - o Allow the interviewee to practice their questions to the interviewer

- Respond: Take notes and give feedback at the end, so that the interviewee can learn from the process. This feedback should:
 - o Clarify what went well and not so well
 - o Present elements of the case that were missed
 - o Direct the interviewee toward areas for improvement
 - o Be comprehensive as you address the interviewee's presences and poise in

addition to the technical skills they present

Note to all interviewers: Always give critical and constructive feedback. Be as honest and forthcoming as will be meaningful to your interviewee. Mutual improvement is your goal and may require uncomfortable conversations.

III. STRATEGIC TOOLS AND FRAMEWORKS

General Tools for Case Interviews

When preparing for case interviews, you will hear repeatedly, “Learn the frameworks,” and “develop the analytical tools.” The problem for most people is that nobody ever explains exactly what frameworks and tools they are talking about. The list presented in the following sections, though not exhaustive, covers many of the standard tools you will rely on when performing case interviews. Use these tools to think about the key issues and to lead you from the facts to a conclusion.

As you become more familiar with these frameworks and tools, you will develop a higher level of comfort in developing your own frameworks (or structures) to tackle a problems. Early on in your practice you will find that you may be reliant on the frameworks to the point where they become your structure. More advanced interviewees can rely on the framework foundations to develop their own structure to solve the problem.

General Tools for Case Interviews

When preparing for case interviews, you will hear repeatedly, “Learn the frameworks,” and “develop the analytical tools.” The problem for most people is that nobody ever explains exactly what frameworks and tools they are talking about. The list presented in the following sections, though not exhaustive, covers many of the standard tools you will rely on when performing case interviews. Use these tools to think about the key issues and to lead you from the facts to a conclusion.

As you become more familiar with these frameworks and tools, you will develop a higher level of comfort in developing your own frameworks (or structures) to tackle a problems. Early on in your practice you will find that you may be reliant on the frameworks to the point where they become your structure. More advanced interviewees can rely on the framework foundations to develop their own structure to solve the problem.